

LECTURE 3 Supplement

Topics

Federal Reserve Banks' Balance Sheet and Monetary Policy

- In the US, the *supply of money* is loosely defined as the the amount of USD available to the public at any given time
- Money supply has two components:
 - USD bank deposits (i.e., checking/savings accounts)
 - USD used by the population in daily transactions, known as *currency in circulation*
- The institution that is in charge of creating/retiring currency is the Federal Reserve Bank (FRB)
- The actual printing of new money is done by the US Treasury Department
- Recall that the FRB is the *Central Bank* of the US (i.e., the bank of the commercial banks in the US)
- In particular, all commercial banks are required to open and maintain a reserve account with the FRB

- Both the currency in circulation and the reserve accounts of commercial banks are considered *liabilities* for the FRB
- On the asset side, FRB holds primarily securities such as US Treasury bonds, but also loans made to US commercial banks or foreign central banks
- The difference between FRB's assets and liabilities represents FRB's *equity*
- This equity includes "shares" that are owned by the US commercial banks (by law, member banks are required to own FRB shares and in exchange receive an attractive dividend)
- The assets, liabilities and the equity make up FRB's *balance sheet*

Federal Reserve Banks
Balance Sheet

Assets	Liabilities
Financial securities:	Currency in circulation (i.e., Federal Reserve notes)
• US Treasuries, • Federal agency debt securities (Fannie Mae, Freddie Mac), • Mortgage-backed securities	Reserve accounts of US commercial banks (i.e., bank deposits at the FRB)
Loans to US commercial banks or foreign central banks	Deposits of foreign central banks at the FRB
Gold certificates (against the US gold reserves held at the US Treasury)	Equity
Coin (actual coins minted by the US Mint – a division of the US Treasury)	Capital accounts • Capital paid in (FRB shares owned by the US commercial banks) • Surplus (excess FRB shares)

- When the population demands more currency than it is available in circulation + reserves, banks will ask the FRB to create more Federal reserve notes (USD currency)
- FRB will then credit the reserve account of the banks with the newly created notes
- In exchange, banks pay the FRB with securities such as US Treasury bonds
- Banks will use the new notes to make loans or fulfill the cash needs of their depositors (the population)
- Effectively, people end up with more cash in hand or in deposits
- Because money supply = currency in circulation + deposits, it has to be the case that *money supply increases*

- An illustration of how money supply changes
- Population demands 100 in additional currency
 - FRB credits banks' reserve accounts by +100
 - Banks pay FRB in securities and FRB is credited +100
- Money supply (= Currency + Reserves) changes by +100

Federal Reserve Banks Balance Sheet	
Assets	Liabilities
Financial securities (+100)	Currency in circulation
Loans	Banks' reserve accounts (+100)
Gold	Other deposits
Coin	
	Equity
	Capital accounts

- The balance sheet of commercial banks also changes
- On the asset side:
 - Bank's reserve account changes by +100
 - Bank's securities changes by -100
- On the liability side: No changes

US Commercial Banks Balance Sheet	
Assets	Liabilities
Financial securities (-100)	Deposits from the population
Cash held in the Reserve account at FRB (+100)	Bonds issued by the banks
Loans	Other liabilities
Other assets	Equity
	Common equity issued by the banks

- Eventually, the cash surplus of 100 will be converted into loans to the population
- The balance sheet of commercial banks becomes:

US Commercial Banks Balance Sheet	
Assets	Liabilities
Financial securities (-100)	Deposits from the population
Cash held in the Reserve account at FRB	Bonds issued by the banks
Loans (+100)	Other liabilities
Other assets	Equity
	Common equity issued by the banks

- The example above illustrates how the money supply increases when the population demands more currency
- More currency in the hands of the population means that people are willing to pay more for the goods that they buy
- The potential increase in the prices of goods across the economy is a phenomenon known as *price inflation*
- Inflation in moderation is a very good thing for an economy: higher prices $\Rightarrow$ higher profits $\Rightarrow$ higher wages $\Rightarrow$ more demand for goods $\Rightarrow$ economy grows $\Rightarrow$ higher prices
- However, notice that the chain of arrows above is self reinforcing: higher prices lead to even higher prices
- If prices increase faster than the economy, people's purchasing power will diminish and the economy will plunge into a recession

- The central bank's goal is to keep the inflation under control such that the economy grows at a sustainable pace
- The only way to control inflation is to control the supply of money available to the population
- As a result, FRB may take actions that have the effect of increasing or decreasing the the supply of money irrespective of the population's demand for currency
- The set of actions that a central bank may take to alter the supply of money available to people is known as *monetary policy*

- A country's monetary policy seeks to achieve a *double mandate*
 - Price stability (maintain a target inflation rate of 2%)
 - Full employment of the workforce
- Full employment means that the economy is growing at a pace that creates enough new jobs for everybody
- Moderate inflation means that prices have to be kept in check so the economy doesn't overheat
- FRB controls the supply of money *indirectly* through the *interest rates* on the reserves of the commercial banks at the FRB

- Banks' reserves at the FRB are subject to three interest rates:
 - *Interest on Reserve Balances (IORB)*: The interest rate that the FRB pays for banks' excess reserves
 - *Discount Window Interest Rate (DWR)*: The interest rate that FRB charges banks that want to borrow directly from FRB (discount window program)
 - *Federal Fund Rate (FFR)*: The interest rate that a bank charges for lending their excess reserves to other banks
- FRB has full control over IORB and DWR
- FRB does not have full control over FFR but it can influence it
- The FFR is determined by the lending and borrowing of reserves among banks (the *interbank* lending market)
 - When reserves are scarce, FFR goes up
 - When reserves are ample, FFR goes down

- The FRB can influence the amount of reserves that banks have, and therefore can influence *indirectly* the FFR
- The FRB affects banks' reserves through buying and selling of securities, known as *open market operations*
- Here's how it works:
 - The FRB buys and sells securities from the banks only
 - When the FRB wants to lower the FFR, it buys securities from the banks
 - To pay for the securities, the FRB credit the banks' reserves in the amount of the value of the securities purchased from the banks
 - Because banks have more reserves, banks borrow less from each other, and the FFR goes down

- Similarly, when the FRB wants to increase the FFR, it sells securities to the banks
- In exchange, the FRB debits the banks' reserves in the amount of the value of the securities sold to the banks
- Because banks have less reserves, banks may have to borrow more from each other, and the FFR goes up
- Thus, when the FRB buys securities, banks' reserves increase
- As a result, money supply (= currency in circulation + reserves) available to the people increases (*expands*)
- Similarly, when the FRB sells securities, banks' reserves decrease and money supply decreases

- The FRB governors meet 8 times a year to decide on the FFR that they want to *target*
- Once they make a decision, the FRB engages in open market operations to implement the target FFR
- For example, if the FRB decided to lower the FFR by 0.5% (the target level), the FRB will start buying securities until the FFR drops to the targeted level
- To furthermore ensure that the FFR stays at the targeted level, the FRB will also lower the other two interest rates on reserves, namely the IORB and DWR
- Similarly, if the FRB decided to increase the FFR, the FRB will start selling securities until the FFR reaches the target level
- To keep the FFR at the targeted level, the FRB will also increase the IORB and DWR

- A country's monetary policy can stir a country out of trouble
- When a country is going through an economic recession, the economy is contracting (i.e., GDP growth rate is negative)
- To jump start the economy, the central bank will use an *expansionary* monetary policy that increases the supply of money available to the people
- As noted above, the central bank can increase the money supply by lowering the target FFR (i.e., central bank is buying securities)
- Similarly, when a country's economic growth pushes prices too high, price inflation can rise dangerously high
- To stem this trend, the central bank will use a *contractionary* monetary policy that reduces the supply of money
- As noted above, the central bank can decrease the money supply by increasing the target FFR (i.e., central bank is selling securities)